Methodology for Union Density in Latin America

Juan M. Campanario, PhD[1]

Introduction

This paper describes the harmonization process used to estimate trade union density rates and median earnings for Latin America disaggregated by sex, age, and broad industry categories. The dashboard presents estimates based on publicly available microdata from national labor force and household surveys for Brazil, Colombia, Guatemala, Mexico, and Peru, which account for over two-thirds of the region's workforce. The International Labour Organization (ILO), through ILOSTAT, and the Organization for Economic Co-operation and Development (OECD), through the OECD/Amsterdam Institute for Advanced Labour Studies (AIAS) Institutional Characteristics of Trade Unions, Wage setting, State Intervention and Social (ICTWSS) also provide some of these statistics at varying degrees of coverage for Latin America. This dashboard's advantages are that it relies solely on national surveys for a larger set of Latin American countries, provides disaggregated estimates by sex, age, and industry, and delivers earnings estimates.[2]

Shown in Table 1, the ILOSTAT series relies on data from a mix of national surveys and administrative records. Brazil, Mexico, and Colombia were estimated from national surveys, while Guatemala and Peru's estimates were derived from administrative records. Deriving all estimates from national surveys is advantageous as it allows for greater comparability across countries. The OECD relies on survey data but does not provide estimates for Peru nor Guatemala.

Table 1: Union Membership – Data Sources

Country	My Source	ILO Source	OECD/AIAS Source
Brazil	Pesquisa Nacional por Amostra de Domicílios Contínua, Visita 1 (PNADC)	PNADC	PNADC
Colombia	Gran Encuesta Integrada de Hogares (GEIH)	Administrative Data	GEIH
Guatemala	Encuesta Nacional de Empleo e Ingresos (ENEI)	ENEI	N/A
Mexico	Encuesta Nacional de Ocupación y Empleo (ENOE)	ENOE	ENOE
Peru	Encuesta Nacional de Hogares (ENAHO)	Administrative Data	N/A

The rest of the document describes the harmonization process, including defining union membership and the target population, comparing earnings estimates with ILOSTAT, developing a consistent industry classification, and handling country-specific limitations.

Defining Target Population & Union Membership Comparison

I define a worker as any individual between 15 and 98 who reported non-zero, positive earnings. I do so because both the ability to join a union and the survey structure for asking about union membership vary by country. Union membership is defined as a binary variable for whether a respondent is a member of a trade union, based on the most relevant question in each survey, typically a question similar to "Are you a member of a union or similar workers' organization?" The exact question asked in each survey is shown below in Table 2. In addition to the survey question itself, each survey has its own methodology and microdata processing pratices which can affect estimation. For example, in Colombia's labor force survey there are no null responses when asked about membership. However, in Guatemala, only workers that responded yes to a related, preceding question were asked about union membership. Thus, the target population

[1] Generative AI (Claude, Anthropic) assisted with improving the data pipeline, code debugging, building the dashboard, and exploratory data analysis. All methodological choices, interpretation, and conclusions are the author's own. For questions, contact: juan.m.campanario@gmail.com

[2] On the dashboard, estimates are suppressed for disaggregated groups with fewer than 200 observations but are available in the accompanying excel files.

was chosen to reflect the limitations, and the estimates provided represent a lower bound for union membership.

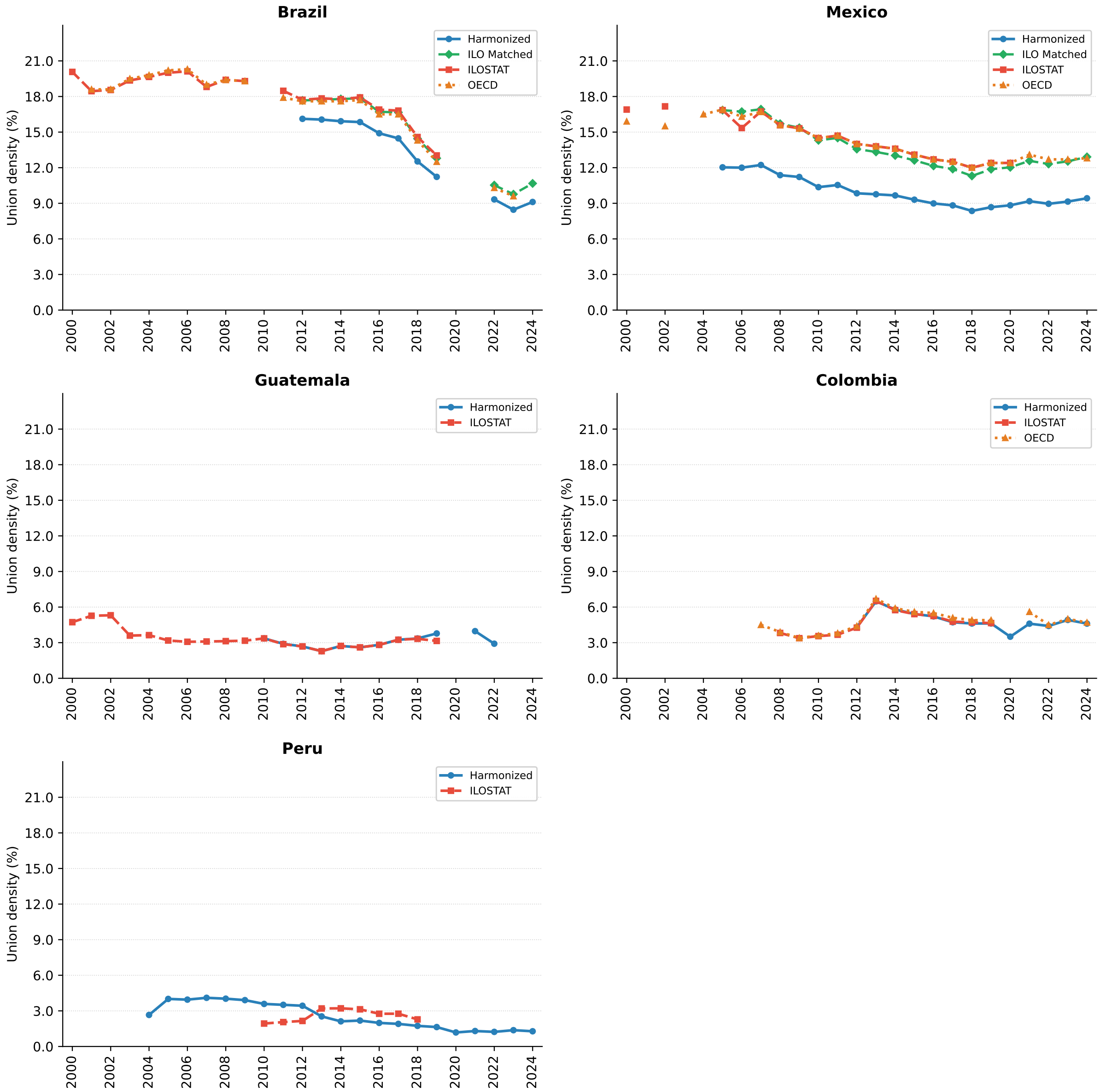
Table 2: Union Membership Survey Questions

Country	**Survey Question**	**English Translation**
Brazil	Na semana de referência era associado a algum sindicato?	During the reference week, were you a member of a trade union?
Colombia	¿Está afiliado o hace parte de una asociación gremial o sindical?	Are you affiliated with or a member of a trade union or workers' organization?
Guatemala	¿En la empresa o institución donde usted trabaja existe sindicato o comité de trabajadores? ¿Está afiliado?	At the company or institution where you work, is there a trade union or workers' committee? Are you affiliated?
Mexico	¿En su empleo ... pertenece a algún sindicato?	At your job, do you belong to a trade union?
Peru	¿Usted o algún miembro de su hogar pertenece o participa en algún grupo, organización y/o asociación: Asociación de trabajadores o sindicato?	Do you or any member of your household belong to or participate in any group, organization, or association: Workers' association or trade union?

Shown in Figure 1, I compare the values published by the OECD and ILOSTAT to the Harmonized estimates. In addition, I plot a series, ILO Matched, in which I attempt to produce estimates similar to the two sources.[3] For Brazil and Mexico, both ILOSTAT and OECD estimates are consistently larger than the Harmonized estimates, but the series follow the same trend for each country. For Brazil, adjusting for the right to unionize (employees excluding military) yields nearly identical results. For Mexico, respondents that were unsure about their status or did not answer are omitted; ILO Matched shows that accounting for this difference yields nearly identical estimates. For Guatemala, imputing a zero for all missing responses yields identical results to ILOSTAT. For Colombia and Peru, ILOSTAT relies on administrative records, but the trends are similar, and both data sources show that the countries have consistently low union membership rates.

[3] OECD/AIAS ICTWSS database's has survey-based union density for Brazil, Colombia, and Mexico, and there are no estimates for Guatemala or Peru.)

Figure 1: Union Rate Methodological Comparison



Note: The Harmonized series defines union membership as a binary variable, with missing values coded as non-members, and the sample is restricted to persons aged 15–98 with positive earnings. ILOSTAT refers to published ILO values, and ILO Matched attempts to replicate the ILO's reported methodology for each country (Brazil and Mexico only). For Colombia and Peru, the ILO draws on administrative records rather than household surveys; no methodological replication is attempted, but estimates are very similar in magnitude. OECD refers to the OECD/AIAS ICTWSS database's survey-based union density, which is not available for Guatemala or Peru.

Earnings Comparison

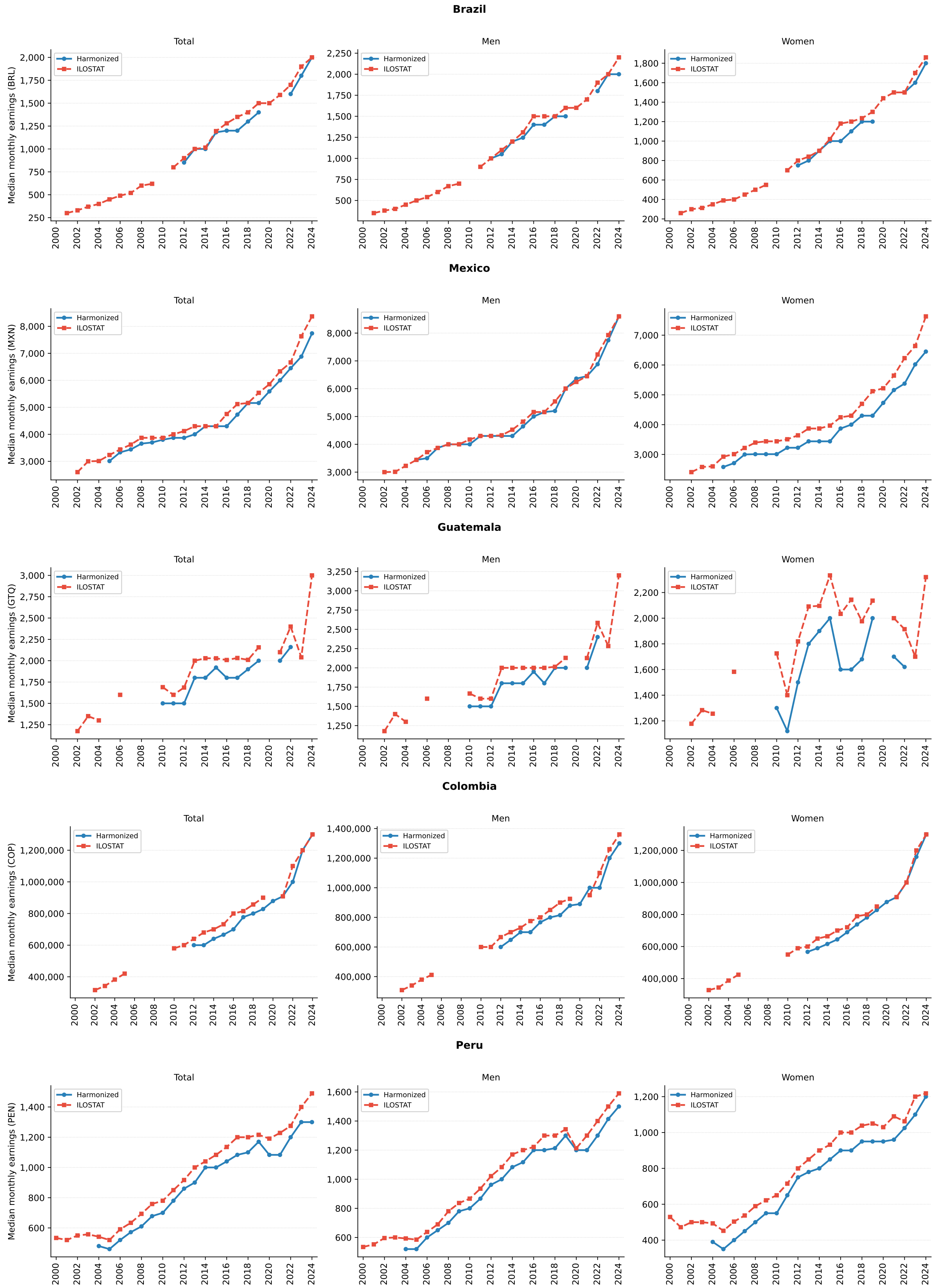
The choice in target population and survey affects comparability of earnings estimates with other sources; a source comparison between the Harmonized series and ILOSTAT is shown in Table 3. In Figure 2, I present the differences in earnings estimates. In the countries for which ILOSTAT calculates median earnings from the same data source, the series are similar in both trend and magnitude. The estimates in the Harmonized series are often lower because the ILO estimates monthly earnings for wage and salary workers, which is a more restrictive and higher-paid group than all persons with positive earnings.

Table 3: Median Monthly Earnings – Data Sources

Country	**Dashboard Source**	**ILOSTAT Source**
Brazil	Pesquisa Nacional por Amostra de Domicílios Contínua, Visita 1 (PNADC)	Pesquisa Nacional por Amostra de Domicílios (PNAD) (2000–2011); PNADC (2012 onwards)
Colombia	Gran Encuesta Integrada de Hogares (GEIH)	Encuesta Continua de Hogares (ECH) (2002–2005); GEIH (2010 onwards)
Guatemala	Encuesta Nacional de Empleo e Ingresos (ENEI)	ENEI
Mexico	Encuesta Nacional de Ocupación y Empleo (ENOE)	ENOE
Peru	Encuesta Nacional de Hogares (ENAHO)	ENAHO (2000–2021); Encuesta Permanente de Empleo (EPE) (2022 onwards)

Note: The Harmonized series for Brazil begins in 2012 (PNADC), so there is no overlap with the earlier PNAD period. For Peru, the Harmonized series continues to use ENAHO post-2022, while ILOSTAT switches to EPE. For all other countries, the Harmonized series uses the same source as ILOSTAT.

Figure 2: Median Earnings Comparison



Note: The Harmonized series produces median monthly earnings in local currency for all persons aged 15–98 with positive earnings from their main job, using survey weights. ILOSTAT reports median monthly earnings of employees (wage and salary workers).

Industry Harmonization

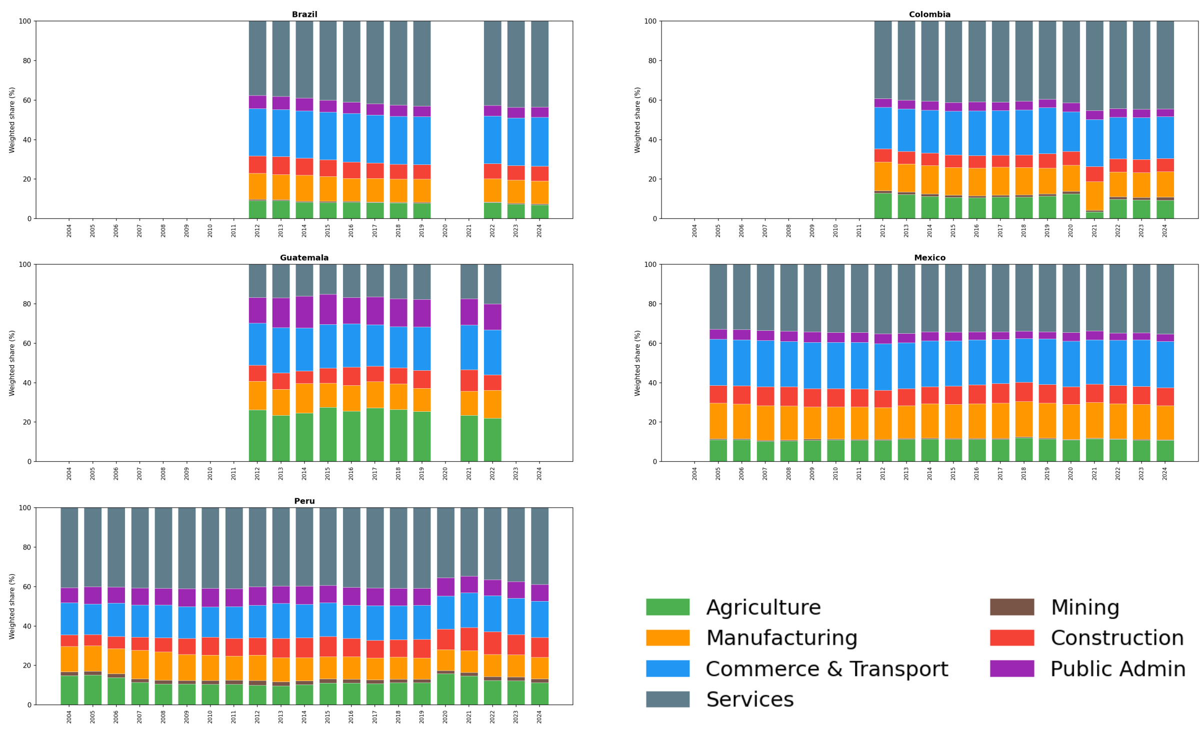
This section documents the construction of a harmonized industry classification for the dashboard. Each country's native survey industry classification is mapped to a common target classification based on the International Standard Industrial Classification of All Economic Activities (ISIC) Rev. 4, aggregated to seven broad sectors. Table 4 contains the sector codes, labels, and the corresponding ISIC sections and division ranges.

Table 4: 7-Sector Classification (ISIC Rev. 4)

Code	**Sector**	**ISIC Section**	**Divisions**
1	Agriculture	A	01–03
2	Mining	B	05–09
3	Manufacturing	C	10–33
4	Construction	F	41–43
5	Commerce & Transport	G–H	45–53
6	Public Administration	O	84
7	Services	All remaining	—

Division 9 (*Mining Support Activities*) is included as part of the Mining sector in all countries where ISIC Rev. 4 or an equivalent is used. The Services sector absorbs all divisions not explicitly assigned to other sectors, including financial services, education, health, and real estate. In Figure 3, the weighted share of in-sample workers by sector for each country and year is plotted to confirm a consistent classification.

Figure 3: Industry Composition



Note: Each bar represents the share of in-sample workers (aged 15–98 with positive earnings), based on availabe survey weights, in each sector annually. Guatemala reports no mining employment, as ENEI lumps Mining and Manufacturing into a single category.

Data Idiosyncrasies and Crosswalk Notes

Table 5: Country Crosswalk Notes

Country	Notes
Brazil	Industry codes are provided using 5-digit ISIC Rev. 4, which are aggregated to the division level.
Colombia	Industry codes are provided for 4-digit ISIC, ISIC Rev. 4 was implemented in 2020, with prior years using ISIC Rev. 3.
Guatemala	Industry codes are provided as pre-aggregated 1-digit sector codes, with Mining and Manufacturing lumped into a single category (assigned as Manufacturing) and Public Administration is lumped with Education and Health; therefore Guatemala's Public Administration and Manufacturing sectors are broader than that for other countries. Industry codes are not available in 2010–2011.
Mexico	Industry codes are provided at the 2 and 4-digit North American Industry Classification System (NAICS), the 2-digit is mapped to ISIC. A small percentage of workers (0.3 percent of the sample) carry an "unspecified" activity code, overwhelmingly paid domestic work in private households.
Peru	Industry codes are coded to ISIC Rev. 4 for all years, through the Updated ENOE.